

1. Smart IoT-driven precision agriculture: Land mapping, crop prediction, and irrigation system

The research paper presents a **Smart IoT-driven Precision Agriculture System** designed to improve agricultural productivity while ensuring sustainable use of resources. The proposed system integrates **land mapping, crop prediction, fertilizer recommendation, and automated irrigation** into a single platform. Using **Landsat-8 satellite images**, the researchers analyzed agricultural land through vegetation and salinity indices to determine land suitability for cultivation. **NDVI (Normalized Difference Vegetation Index)** and **K-means clustering** were used to identify healthy vegetation areas, while **NDSI (Normalized Difference Salinity Index)** helped detect saline regions that could negatively affect crop growth. In addition, an **LSTM deep learning model** was employed to forecast future vegetation conditions, supporting long-term agricultural planning.

The study also developed an **IoT-based crop recommendation system** that collected real-time soil information using sensors measuring nitrogen, phosphorus, potassium (NPK), soil moisture, pH, temperature, and humidity. Several machine learning algorithms were tested, and the **Random Forest model achieved the highest crop prediction accuracy of 97.35%**, making it the most effective approach for selecting suitable crops. Furthermore, the system provided **fertilizer recommendations** based on the nutrient requirements of the soil, helping farmers apply fertilizers more efficiently. To optimize water usage, the researchers designed a **solar-powered irrigation system controlled by fuzzy logic**. Experimental results showed that this intelligent irrigation approach significantly reduced unnecessary water consumption, saving approximately **61–70% more water** compared to conventional irrigation methods. Overall, the research demonstrates that combining **IoT, machine learning, remote sensing, and fuzzy logic** can enhance decision-making in agriculture, increase crop yield, conserve resources, and promote sustainable farming practices.